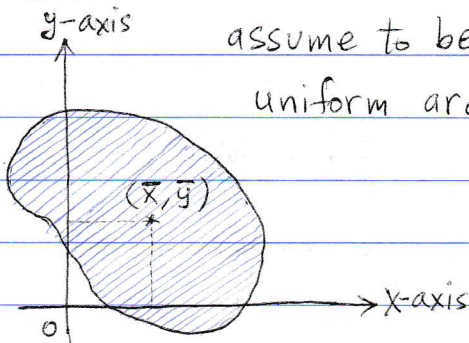


CENTRE OF MASS AND CENTROID

A Plane lamina: A lamina is a region in the x - y plane, which we assume to be a thin sheet of material with uniform area density ρ . (ρ is measured in kg/m^2)
 (*We usually set $\rho = 1 \text{ kg/m}^2$ *)



The centre of mass is a pivot point at which the lamina would exactly balance.

CENTER OF MASS: We define the centre of mass $(\bar{x}, \bar{y})$ of the lamina by

$$\bar{x} = \frac{M_y}{M} = \frac{\rho \int_a^b x f(x) dx}{\rho \int_a^b f(x) dx} = \frac{\int_a^b x f(x) dx}{\int_a^b f(x) dx}$$

$$\bar{y} = \frac{M_x}{M} = \frac{\rho \int_a^b \frac{1}{2} [f(x)]^2 dx}{\rho \int_a^b f(x) dx} = \frac{\frac{1}{2} \int_a^b [f(x)]^2 dx}{\int_a^b f(x) dx}$$

*If the lamina has constant density then the center of mass is called CENTROID of the region.

FIRST MOMENTS

$$M_y = \rho \int_a^b x f(x) dx \quad (\text{Around } y\text{-axis})$$

$$M_x = \rho \int_a^b \frac{1}{2} [f(x)]^2 dx \quad (\text{Around the } x\text{-axis})$$

CENTROID OF A REGION (LAMINA) WITH CONSTANT DENSITY

$$\bar{x} = \frac{M_y}{\rho A}, \quad \bar{y} = \frac{M_x}{\rho A}, \quad \text{where } A = \int_a^b f(x) dx \leftarrow \text{Area of the region/lamina}$$

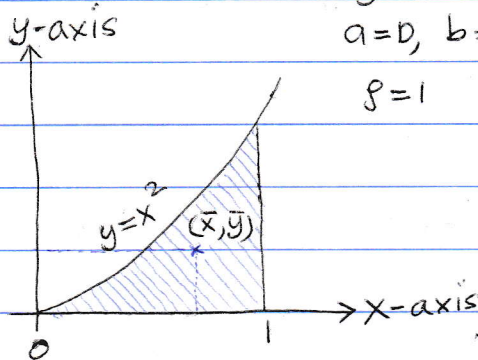
$M_x, M_y \leftarrow 1^{\text{st}}$ moments about

Example 1

Calculate the centroid of the region in the first quadrant bounded by the curve $y=x^2$, the x-axis and the line $x=1$.

Solution

sketch



Here $y=f(x)=x^2$
 $a=0, b=1$
 $g=1$

First Moments

$$M_y = \int_0^1 x f(x) dx = \int_0^1 x^3 dx = \left[\frac{x^4}{4} \right]_0^1 = \frac{1}{4}$$

$$M_x = \frac{1}{2} \int_0^1 [f(x)]^2 dx = \frac{1}{2} \int_0^1 x^4 dx = \frac{1}{2} \left[\frac{x^5}{5} \right]_0^1 = \frac{1}{10}$$

$$\text{Area: } A = \int_0^1 f(x) dx = \int_0^1 x^2 dx = \left[\frac{1}{3} x^3 \right]_0^1 = \frac{1}{3}$$

* Centroid is $(\bar{x}, \bar{y})$ where:

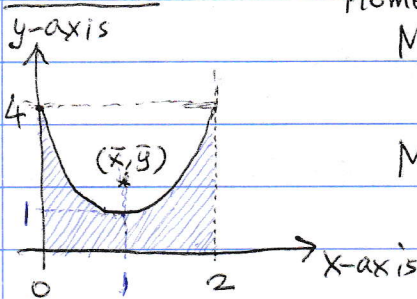
$$\bar{x} = \frac{M_y}{A} = \frac{1/4}{1/3} = \frac{3}{4} ; \quad \bar{y} = \frac{M_x}{A} = \frac{1/10}{1/3} = \frac{3}{10} \quad \therefore \left(\frac{3}{4}, \frac{3}{10} \right)$$

NB: The centroid lies inside lamina

Example 2

Find the centroid of the region bounded by $y=3x^2-6x+4$ the x-axis, y-axis and the line $x=2$.

Solution



Moments: 2

$$M_y = \int_0^2 x f(x) dx = \int_0^2 (3x^3 - 6x^2 + 4x) dx = \left[\frac{3x^4}{4} - 2x^3 + 2x^2 \right]_0^2 = 4$$

$$M_x = \frac{1}{2} \int_0^2 [f(x)]^2 dx = \frac{1}{2} \int_0^2 (9x^4 - 36x^3 + 60x^2 - 48x + 16) dx$$

$$= \frac{1}{2} \left[\frac{9x^5}{5} - 9x^4 + 20x^3 - 24x^2 + 16x \right]_0^2 = \frac{24}{5}$$

$$\text{Area: } A = \int_0^2 f(x) dx = \int_0^2 (3x^2 - 6x + 4) dx = \left[\frac{3x^3}{3} - 3x^2 + 4x \right]_0^2 = 4$$

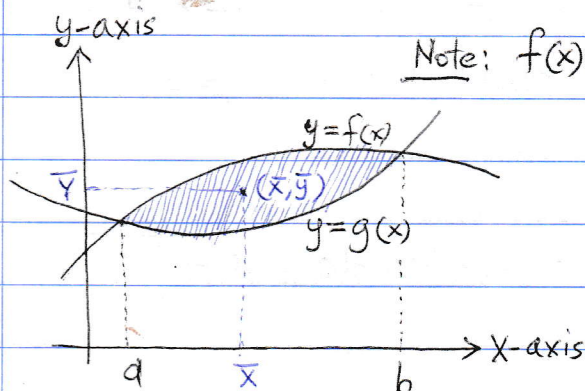
Centroid

$$\bar{x} = \frac{M_y}{A} = \frac{4}{4} = 1 ; \quad \bar{y} = \frac{M_x}{A} = \frac{24/5}{4} = \frac{24}{20} = \frac{6}{5}$$

$\therefore$ Centroid $\left(1, \frac{6}{5} \right)$

NB: Here centroid lies outside the lamina.

CENTROID OF A REGION BY TWO CURVES



Note: $f(x) > g(x)$ on interval $x \in (a, b)$.

* First Moments

$$M_y = \rho \int_a^b x [f(x) - g(x)] dx$$

$$M_x = \frac{\rho}{2} \int_a^b \{ [f(x)]^2 - [g(x)]^2 \} dx$$

Centroid: $(\bar{x}, \bar{y})$

Area: $A = \int_a^b [f(x) - g(x)] dx$

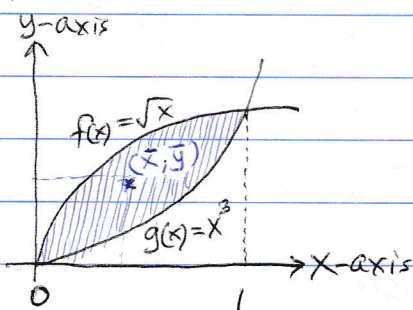
$$\bar{x} = \frac{M_y}{\rho A} ; \quad \bar{y} = \frac{M_x}{\rho A} \quad \boxed{\text{set } \rho = 1}$$

Example 3

Find the centroid of the region bounded by curves $y = x^3$ and $y = \sqrt{x}$.

Solution

Sketch



Solve: $f(x) = g(x) \Leftrightarrow \sqrt{x} = x^3$

$x(1-x^5) = x - x^6 = 0$ ie $\underline{x=0}$, $\underline{x=1}$

First Moments

$$M_y = \int_0^1 x (\sqrt{x} - x^3) dx = \int_0^1 (x^{3/2} - x^4) dx$$

$$= \left[\frac{2}{5} x^{5/2} - \frac{x^5}{5} \right] \Big|_0^1 = \frac{2}{5} - \frac{1}{5} = \frac{1}{5}$$

$$M_x = \frac{1}{2} \int_0^1 [(\sqrt{x})^2 - (x^3)^2] dx = \frac{1}{2} \int_0^1 (x - x^6) dx$$

Area: $A = \int_0^1 (\sqrt{x} - x^3) dx$

$$= \left[\frac{2}{3} x^{3/2} - \frac{x^4}{4} \right] \Big|_0^1 = \frac{2}{3} - \frac{1}{4} = \frac{5}{12}$$

$$= \frac{1}{2} \left[\frac{x^2}{2} - \frac{x^7}{7} \right] \Big|_0^1 = \frac{1}{2} \left(\frac{1}{2} - \frac{1}{7} \right) = \frac{5}{28}$$

Centroid:

$$\bar{x} = \frac{M_y}{A} = \frac{1/5}{5/12} = \frac{12}{25} ; \quad \bar{y} = \frac{M_x}{A} = \frac{5/28}{5/12} = \frac{12}{28} = \frac{3}{7}$$

$\therefore$ Centroid $\left(\frac{12}{25}, \frac{3}{7} \right)$

Theorem (Symmetry theorem for finding the centroid.)

* If the lamina is symmetric about the line $x=c$, then $\bar{x}=c$.

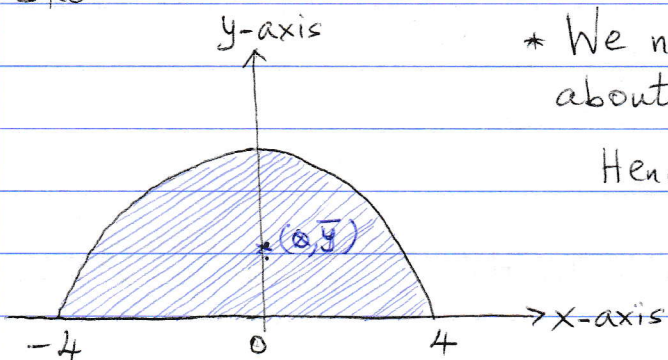
** If the lamina is symmetric about the line $y=d$, then $\bar{y}=d$.

Example 4

Calculate the centroid of the semicircle $y=\sqrt{16-x^2}$.

Solution:

Sketch:



* We note, the semi-circle is symmetric about the line $x=0$.

$$\text{Hence } \bar{x} = \frac{M_y}{A} = 0.$$

* First Moment abt x:

$$M_x = \frac{1}{2} \int_{-4}^4 [f(x)]^2 dx = \frac{1}{2} \int_{-4}^4 (16-x^2) dx = \frac{1}{2} \left[16x - \frac{1}{3}x^3 \right] \Big|_{-4}^4$$
$$= \frac{1}{2} \left[64 - \frac{64}{3} - \left(-64 + \frac{64}{3} \right) \right]$$

$$\therefore M_x = \frac{128}{3}$$

* Area:

$$A = \frac{1}{2} \pi r^2 = \frac{1}{2} \pi (4)^2 = 8\pi$$

$$\bar{y} = \frac{M_x}{A} = \frac{128}{3} \div 8\pi = \frac{128}{3} \times \frac{1}{8\pi} = \frac{16}{3\pi}$$

$$\therefore \text{Centroid } \left(0, \frac{16}{3\pi} \right)$$